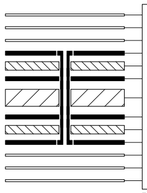
Layer Stack Legend



Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen		Direct Printing	Legend	GBR
	F.Mask	0,02mm	Solder Resist	Solder Mask	GBR
Copper	**L1 (Sig. GND)**	**0,07mm (2,00oz)**		**Signal**	**GBR**
Prepreg		0,1mm	FR4_7628	Dielectric	
Copper	**L2 (GND)**	**0,035mm (1,00oz)**		**Plane**	**GBR**
Core		1,21mm	FR4	Dielectric	
Copper	**L3 (PWR)**	**0,035mm (1,00oz)**		**Plane**	**GBR**
Prepreg		0,1mm	FR4_7628	Dielectric	
Copper	**L4 (Sig. GND)**	**0,07mm (2,00oz)**		**Signal**	**GBR**
	B.Mask	0,02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen		Direct Printing	Legend	GBR
	B.Paste			Paste Mask	

Total thickness: 1,66mm
Note: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	50	±10 %	L1	0,2032	0,28	L2

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

4) SURFACE FINISH: IMMERSION GOLD

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLACK.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING YELLOW NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

9) PCB MATERIAL REQUIREMENTS:

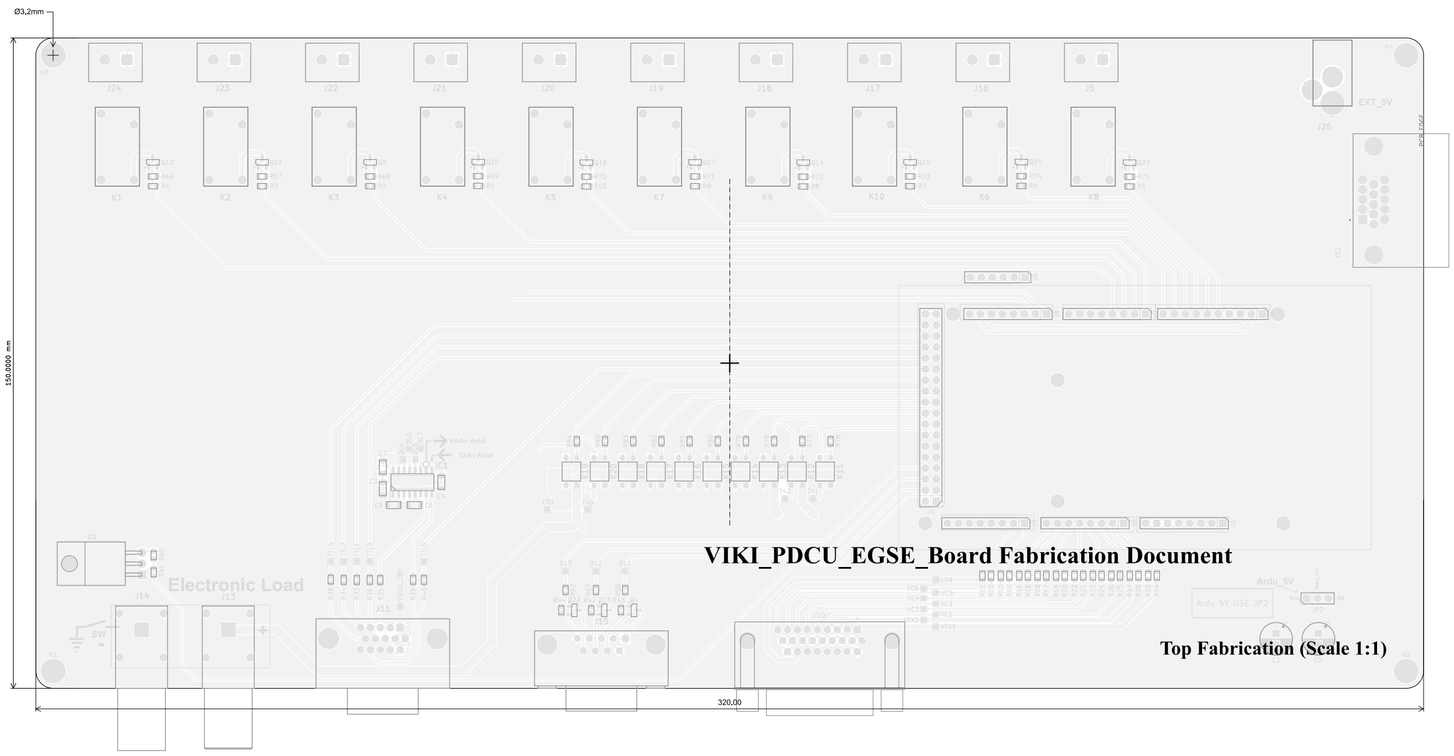
   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY REALTRA SPACE SYSTEMS LIMITED.

10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	320,000 × 150,000 mm
BOARD THICKNESS	1,660 mm
TRACE WIDTH	0,200 mm
TRACE TO TRACE	0,200 mm
MIN. HOLE (PTH)	0,250 mm
MIN. HOLE (NPTH)	3,200 mm
ANNULAR RING	0,150 mm
COPPER TO HOLE	0,254 mm
COPPER TO EDGE	0,250 mm
HOLE TO HOLE	0,254 mm

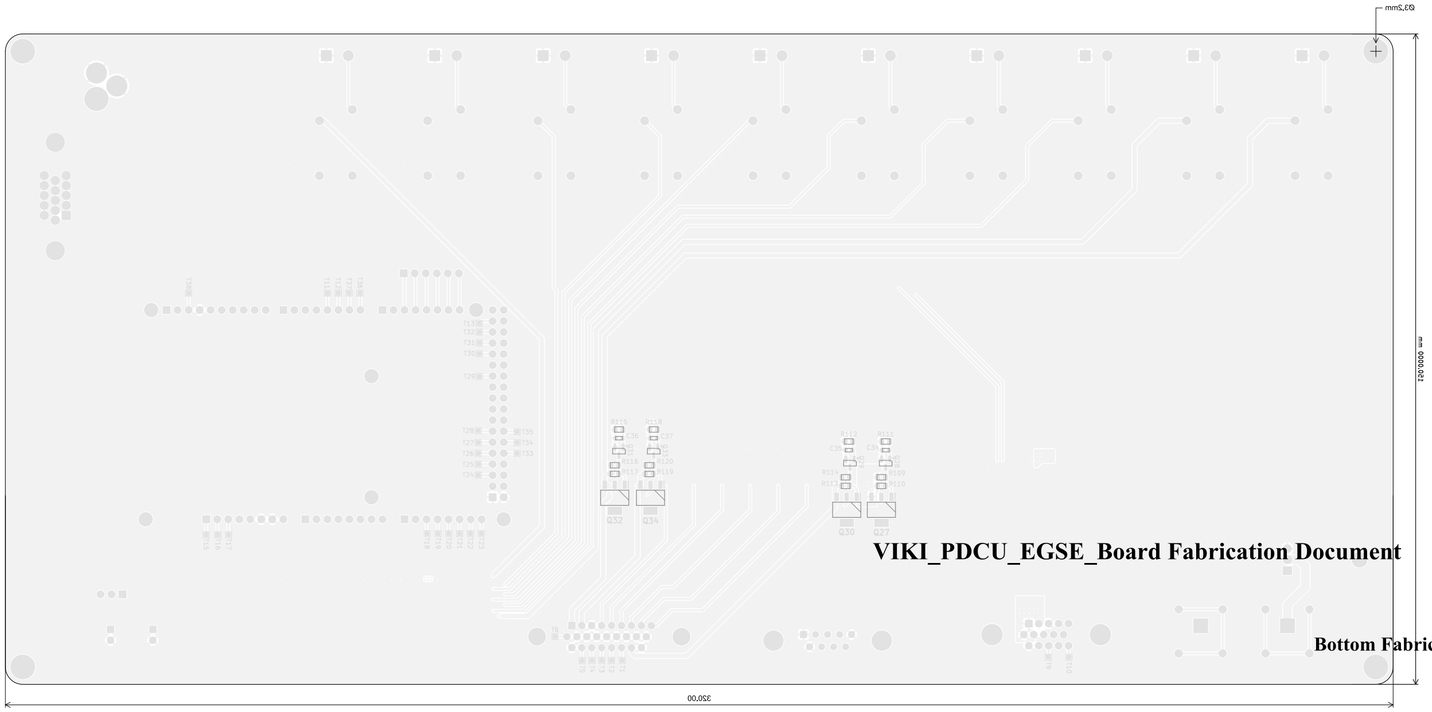
11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.

12) CONFIRM SPACE WIDTHS AND SPACINGS.



All dimensions are in millimeters unless otherwise specified.

Comments:	Company: REALTRA SPACE SYSTEMS LIMITED		Variant: CHECKED	Git Hash: a2a6e93
	Board Name: VIKI_PDCU_EGSE_Board		Project Name: VIKI_PDCU_EGSE	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: VPDCU_EGSE.kicad_pcb	Designer: Leslie Pen	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A2	Sheet: 1 of 10



VIKI_PDCU_EGSE_Board Fabrication Document

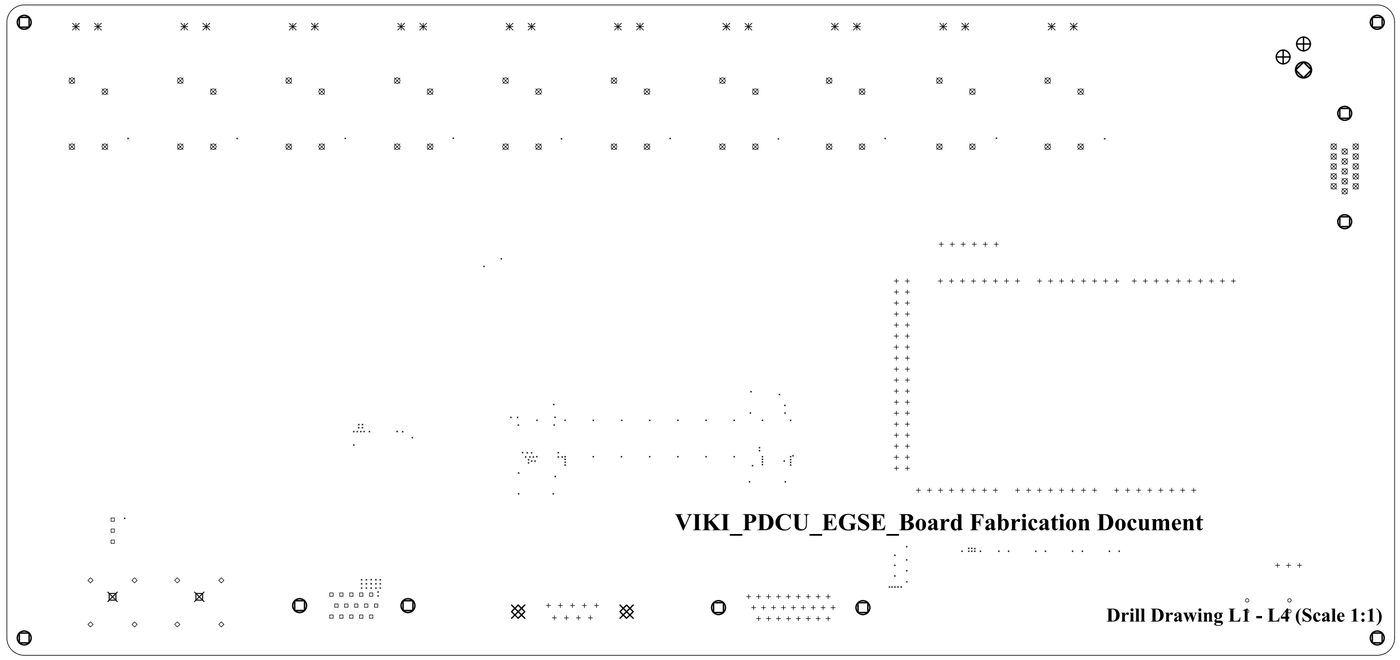
Bottom Fabrication (Scale 1:1)

All dimensions are in millimeters unless otherwise specified.

Comments:	Company: REALTRA SPACE SYSTEMS LIMITED		Variant: CHECKED	Git Hash: a2a6e93
	Board Name: VIKI_PDCU_EGSE_Board		Project Name: VIKI_PDCU_EGSE	
Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: VPDCU_EGSE.kicad_pcb	Designer: Leslie Pen	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:	Reviewer:	Size: A2	Sheet: 2 of 10

Drill Table

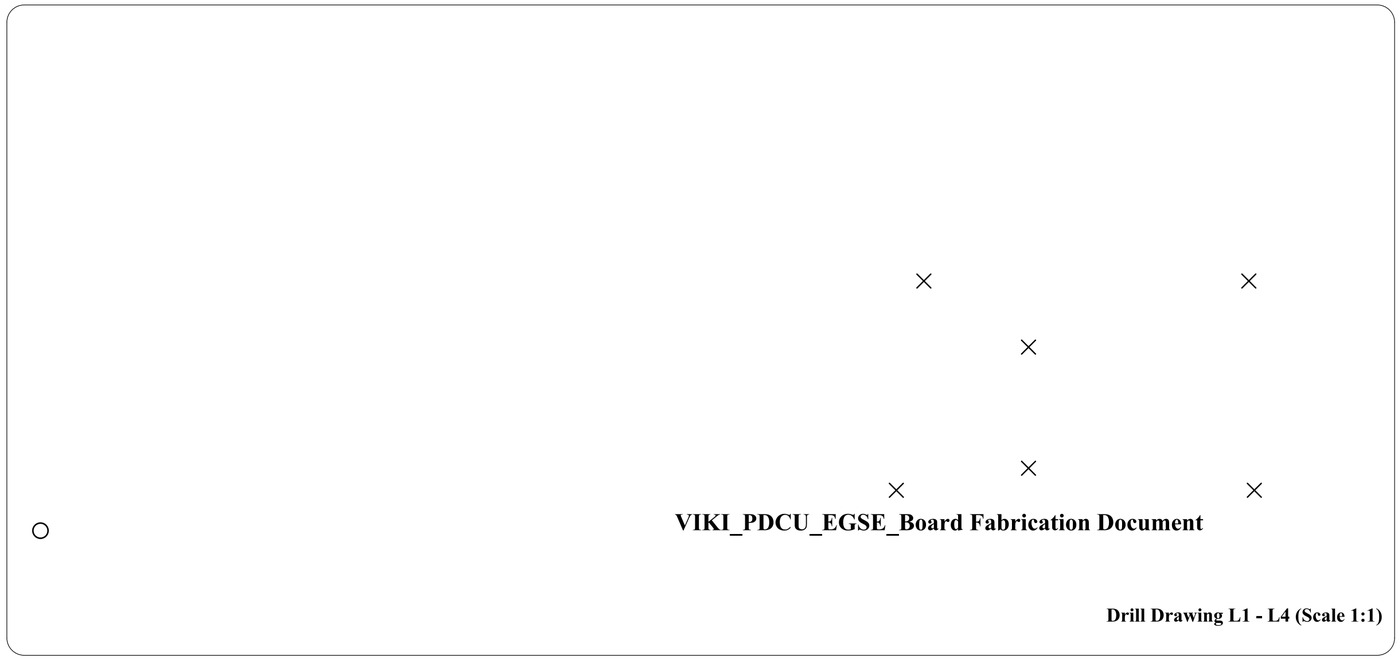
Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	130	0,25mm (9,84mils)	PTH	Round	L1 (Sig, GND) - L4 (Sig, GND)	Via
○	4	0,80mm (31,50mils)	PTH	Round	L1 (Sig, GND) - L4 (Sig, GND)	Pad
+	130	1,00mm (39,37mils)	PTH	Round	L1 (Sig, GND) - L4 (Sig, GND)	Pad
□	18	1,10mm (43,31mils)	PTH	Round	L1 (Sig, GND) - L4 (Sig, GND)	Pad
◇	8	1,20mm (47,24mils)	PTH	Round	L1 (Sig, GND) - L4 (Sig, GND)	Pad
⊠	55	1,30mm (51,18mils)	PTH	Round	L1 (Sig, GND) - L4 (Sig, GND)	Pad
✳	20	1,60mm (62,99mils)	PTH	Round	L1 (Sig, GND) - L4 (Sig, GND)	Pad
✳	2	2,20mm (86,61mils)	PTH	Round	L1 (Sig, GND) - L4 (Sig, GND)	Pad
⊠	2	3,05mm (120,08mils)	PTH	Round	L1 (Sig, GND) - L4 (Sig, GND)	Pad
⊕	2	3,16mm (124,50mils)	PTH	Round	L1 (Sig, GND) - L4 (Sig, GND)	Pad
○	10	3,20mm (125,98mils)	PTH	Round	L1 (Sig, GND) - L4 (Sig, GND)	Pad
○	1	3,64mm (143,31mils)	PTH	Round	L1 (Sig, GND) - L4 (Sig, GND)	Pad
	Total: 587					



Comments:	Company: REALTRA SPACE SYSTEMS LIMITED		Variant: CHECKED	Git Hash: a2a6e93
	Board Name: VIKI_PDCU_EGSE_Board		Project Name: VIKI_PDCU_EGSE	
Sheet Title: Drill Drawing (L1 - L4)	File Name: VPDCU_EGSE.kicad_pcb	Designer: Leslie Pen	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A2	Sheet: 3 of 10

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	6	3,20mm (125,98mils)	NPTH	Round	L1 (Sig, GND) - L4 (Sig, GND)	Mechanical
○	1	3,50mm (137,80mils)	NPTH	Round	L1 (Sig, GND) - L4 (Sig, GND)	Mechanical
	Total 7					



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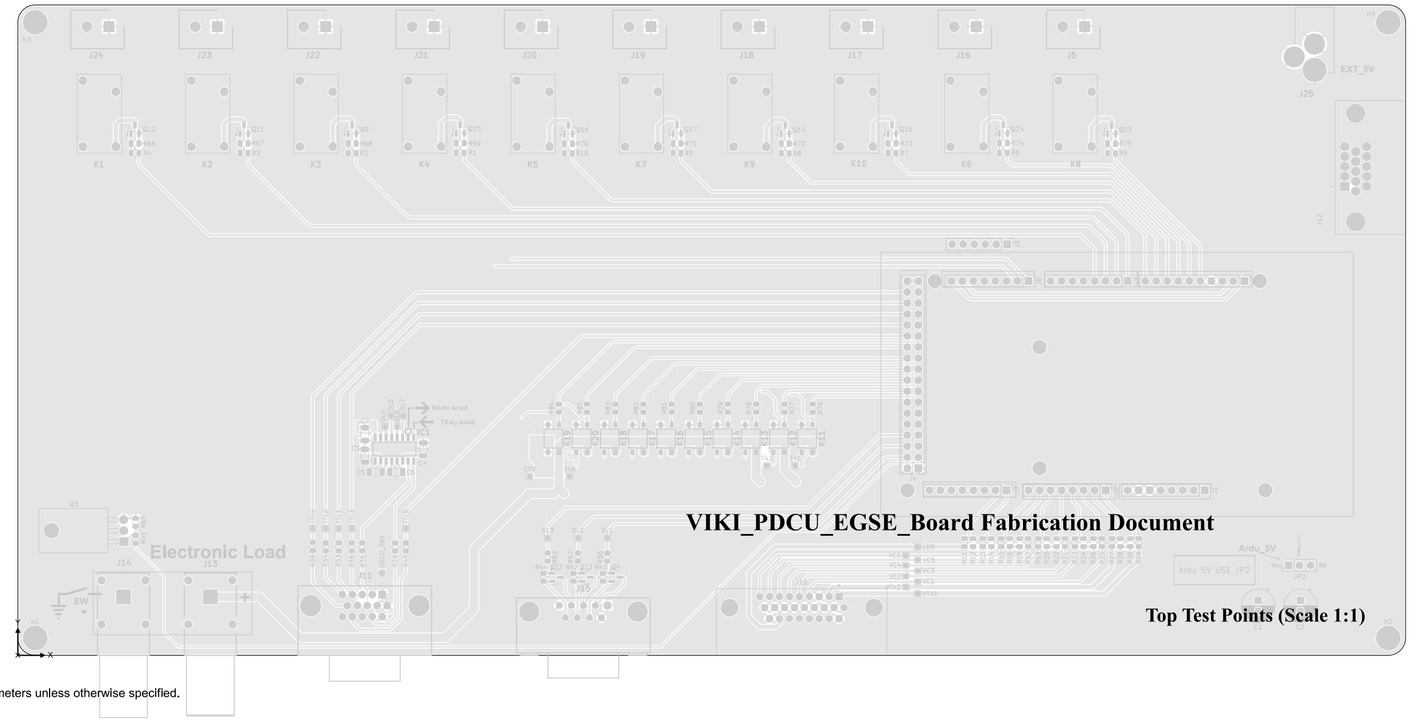
Sheet Title: Drill Drawing (L1 - L4)	File Name: VPDCU_EGSE.kicad_pcb	Designer: Leslie Pen	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A2	Sheet: 4 of 10

Ref.	Net	X [mm]	Y [mm]

Ref.	Net	X [mm]	Y [mm]



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Comments:	Company: REALTRA SPACE SYSTEMS LIMITED	Variant: CHECKED	Git Hash: a2a6e93
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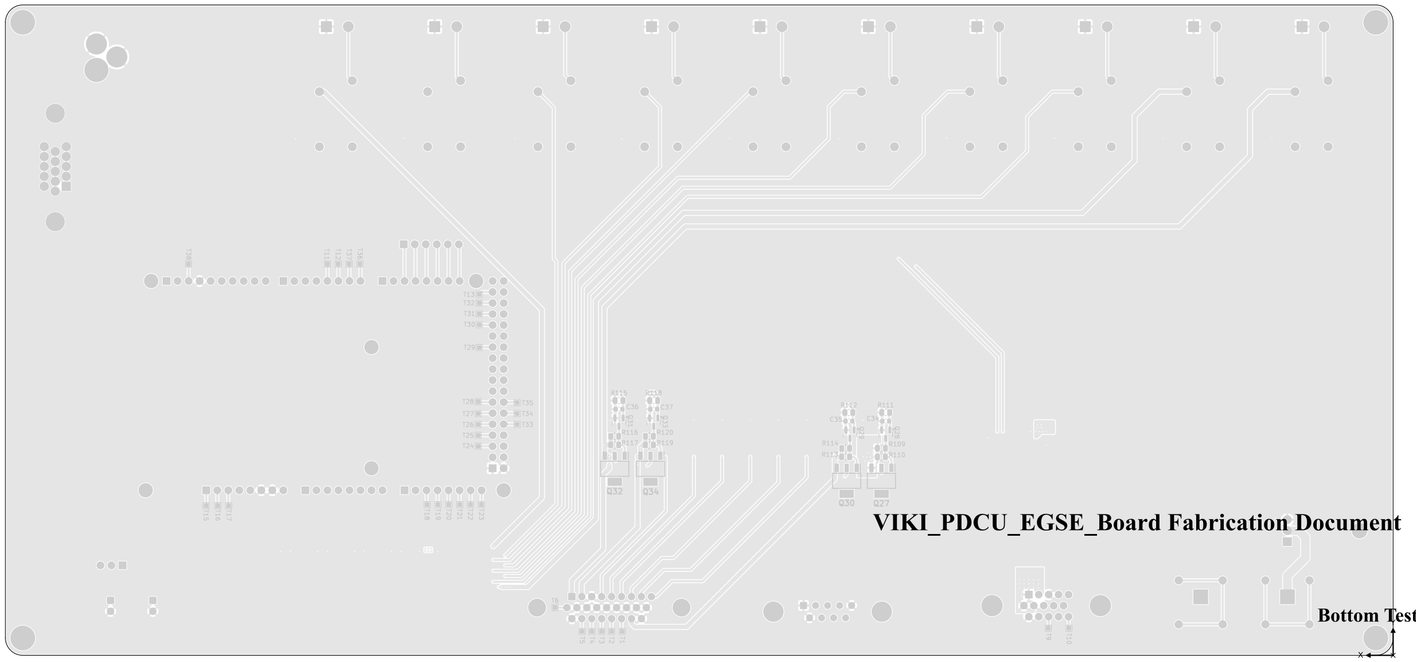
Sheet Title: Top Test Points (Scale 1:1)	File Name: VPDCU_EGSE.kicad_pcb	Designer: Leslie Pen	Date: 2024-04-13 / Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A2 / Sheet: 5 of 10

Ref.	Net	X [mm]	Y [mm]



All dimensions are in millimeters unless otherwise specified.

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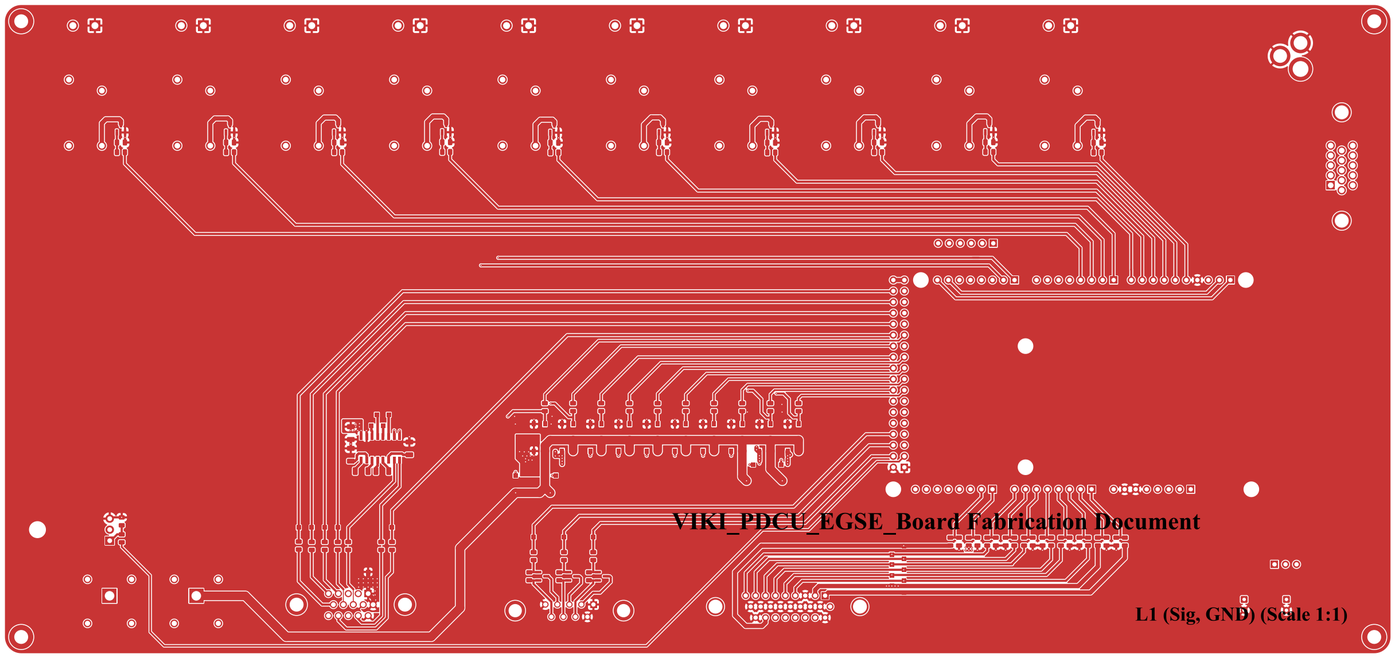
Sheet Title: Bottom Test Points (Scale 1:1)	File Name: VPDCU_EGSE.kicad_pcb	Designer: Leslie Pen	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A2	Sheet: 6 of 10



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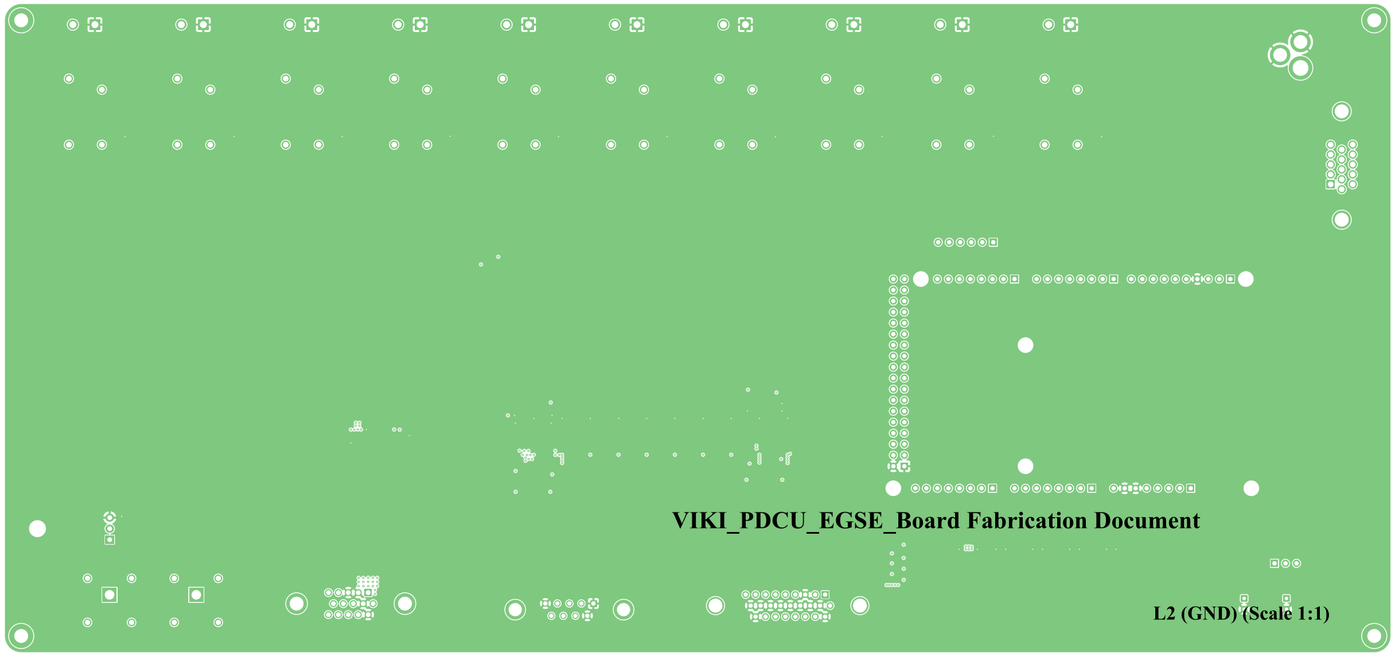
Sheet Title: L1 (Sig, GND) (Scale 1:1)	File Name: VPDCU_EGSE.kicad_pcb	Designer: Leslie Pen	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A2	Sheet: 7 of 10



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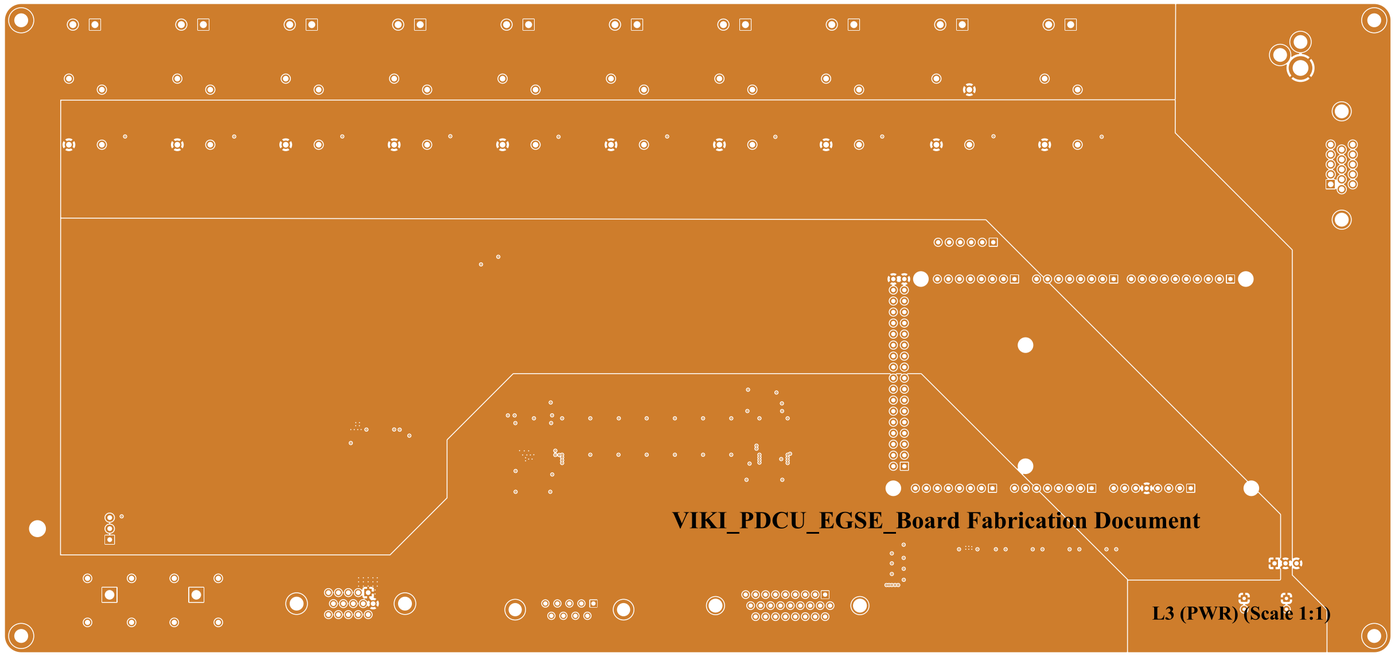
Sheet Title: L2 (GND) (Scale 1:1)	File Name: VPDCU_EGSE.kicad_pcb	Designer: Leslie Pen	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A2	Sheet: 8 of 10



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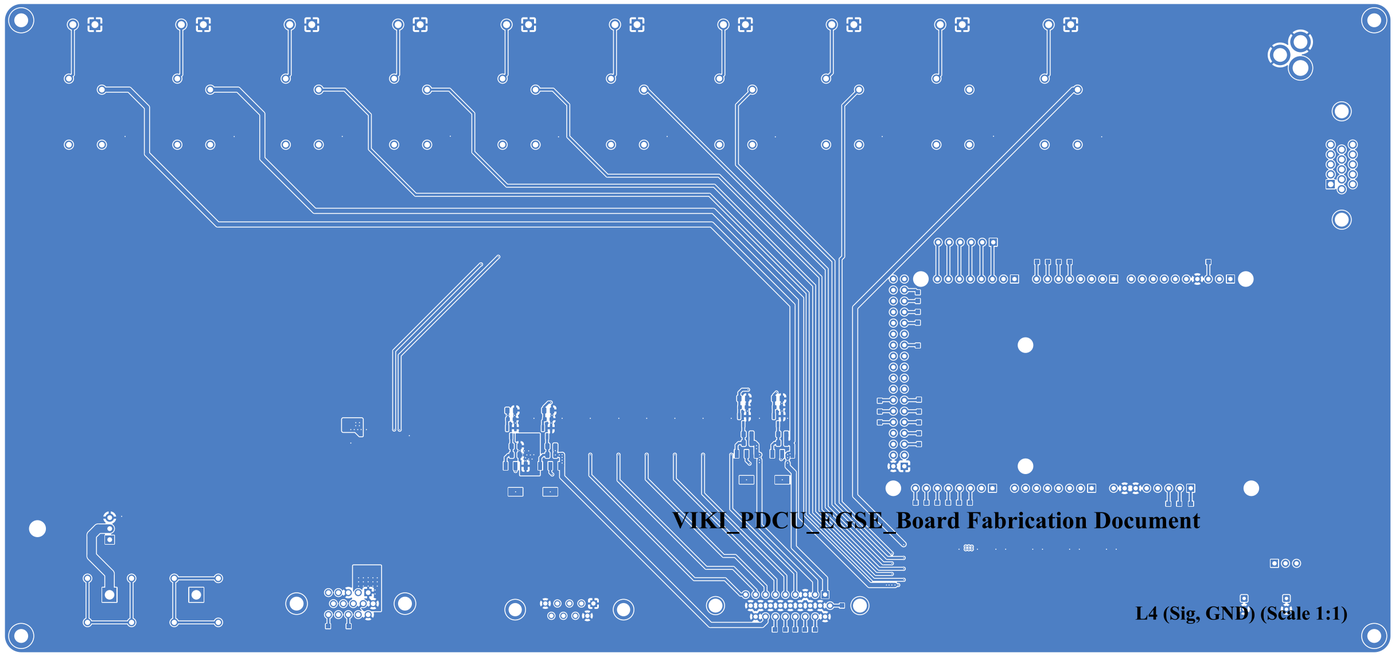
	Board Name: VIKI_PDCU_EGSE_Board		Project Name: VIKI_PDCU_EGSE	
Sheet Title: L3 (PWR) (Scale 1:1)	File Name: VPDCU_EGSE.kicad_pcb	Designer: Leslie Pen	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A2	Sheet: 9 of 10



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Sheet Title: L4 (Sig, GND) (Scale 1:1)	File Name: VPDCU_EGSE.kicad_pcb	Designer: Leslie Pen	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A2	Sheet: 10 of 10